# Develop AI Agents on Azure

## Agents, tools, orchestration, and interoperability

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# What You'll Learn

- Explain what turns a chat model into an agent
- Give an agent tools, and decide who is allowed to run them
- Connect an agent to company knowledge and to Microsoft 365
- Choose an orchestration shape when one agent is not enough
- Recognise when to use MCP and when to use A2A

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# Problem: A Chat App Cannot Do Anything

A chat model produces text. Ask it to "book the meeting room" and it produces a *description* of booking a meeting room.

- ❌ It cannot look up today's data
- ❌ It cannot call your systems
- ❌ It stops after one answer, even if the job needs five steps

**Need:** something that can think, act, look at the result, and keep going until the job is done.

**Agent = model + instructions + tools + a loop.**

The loop is the part people forget. It is the whole trick.



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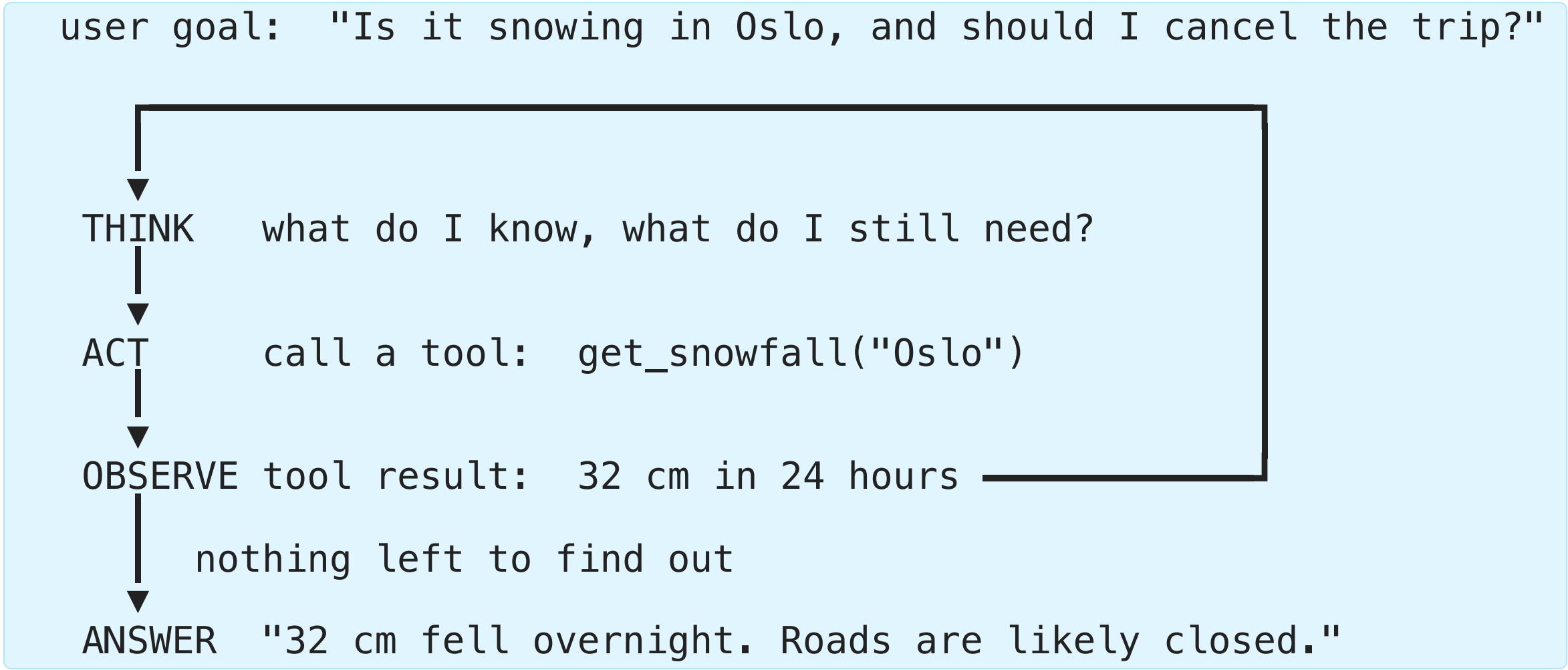
# The Agent Loop




**The model decides when to stop looping.** Instructions, tool descriptions, and approval settings are how you keep that decision under control.

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# Agent Service

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# What is an AI Agent?

***[EXAM]***

**Problem:** LLMs respond to a single prompt; they cannot take multi-step actions

**Agent = model + instructions + tools + automatic tool calling**

**Agent types in Microsoft Foundry:**

| Type | Sub-type | Description |
|---|---|---|
| **Declarative** | Prompt-based | Model + instructions + tools + prompts (most common) |
| **Declarative** | Workflow | Multi-agent orchestration defined in YAML |
| **Hosted** | — | Containerized code-based agents hosted by Foundry platform |

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# Agent Service Features

| Feature | Description |
|---|---|
| **Automatic tool calling** | Agent decides when and how to invoke tools |
| **Securely managed data** | Conversation data managed via Responses API |
| **Extensive tool catalog** | Functions, code interpreter, file search, MCP servers |
| **Enterprise security** | Keyless auth, content safety integration |
| **Customizable storage** | Platform-managed or bring your own Azure Blob |
| **Observability/tracing** | Built-in trace logging |
| **Model selection** | Any deployed model |

**Required resources:** Foundry project + model deployment
**Optional:** Azure AI Search, Azure Storage, Azure Key Vault, Azure Functions



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# Development Approaches

| Approach | Best for |
| --- | --- |
| **Foundry portal** | No-code visual design, playground testing |
| **VS Code extension (Microsoft Foundry)** | Code-first, YAML, version control |

**VS Code Agent YAML pattern:**

```yaml
# yaml-language-server: $schema=https://aka.ms/ai-foundry-vsc/agent/1.0.0
version: 1.0.0
name: my-agent
model:
  id: "gpt-4.1"
  options:
    temperature: 0.5
    top_p: 1
instructions: |
  You are a helpful assistant.
tools: []
```

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# Create Agent via SDK

***[EXAM]***

```python
from azure.ai.projects import AIProjectClient
from azure.ai.projects.models import PromptAgentDefinition

project_client = AIProjectClient(
    credential=DefaultAzureCredential(),
    endpoint="https://{resource-name}.services.ai.azure.com/api/projects/{project-name}"
)

agent = project_client.agents.create_version(
    name="my-agent",
    definition=PromptAgentDefinition(
        model="gpt-4.1",
        instructions="You are a helpful assistant.",
        tools=[],
    )
)

agent = project_client.agents.get(name="my-agent")   # retrieve an existing agent
```

**Keyless access:** `DefaultAzureCredential()` plus the **`get`** operation retrieves an existing agent through Entra ID. An API-key credential is not Entra ID, and `create` or `delete` do not return the existing agent.

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# Custom Tools

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# Retrieval Practice — Lab 1

Pause here and complete Lab 1 before continuing.

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# Custom Tool Options

***[EXAM]***

| Option | Description |
| --- | --- |
| **Custom function** (function calling) | Describe structure; agent decides when to invoke |
| **Azure Functions** | Event-driven via queue triggers; input/output bindings |
| **OpenAPI specification** | Connect to any HTTP API via OpenAPI 3.0 spec |
| **Azure Logic Apps** | Low-code/no-code workflow integration |

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# Retrieval Practice — Lab 2

Pause here and complete Lab 2, then check its solution.

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# Function Calling

```python
from azure.ai.projects.models import FunctionTool, PromptAgentDefinition

function_tool = FunctionTool(
    name="recent_snowfall",
    parameters={
        "type": "object",
        "properties": {
            "location": {"type": "string", "description": "City name"},
        },
        "required": ["location"],
        "additionalProperties": False
    },
    description="Get recent snowfall totals for a given location.",
    strict=True,
)

agent = project_client.agents.create_version(
    name="snowfall-agent",
    definition=PromptAgentDefinition(
        model="gpt-4.1",
        instructions="Help users find snowfall data.",
        tools=[function_tool],
    )
)
```

**`strict=True`** enforces the parameter schema

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# Azure Functions Tool

```python
from azure.ai.projects.models import (
    AzureFunctionTool, AzureFunctionDefinition,
    AzureFunctionBinding, AzureFunctionStorageQueue
)

tool = AzureFunctionTool(
    azure_function=AzureFunctionDefinition(
        input_binding=AzureFunctionBinding(
            storage_queue=AzureFunctionStorageQueue(
                queue_name="INPUT_QUEUE",
                queue_service_endpoint="STORAGE_QUEUE_ENDPOINT",
            )
        ),
        output_binding=AzureFunctionBinding(
            storage_queue=AzureFunctionStorageQueue(
                queue_name="OUTPUT_QUEUE",
                queue_service_endpoint="STORAGE_QUEUE_ENDPOINT",
            )
        ),
        function=AzureFunctionDefinitionFunction(
            name="queue_trigger",
            description="Process a request",
            parameters={...},
        ),
    )
)
```

**Communication via Storage Queue** (input + output bindings)

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# OpenAPI Tool

## Supports OpenAPI 3.0 only

```
from azure.ai.projects.models import OpenApiTool, OpenApiAnonymousAuthDetails

tool = OpenApiTool(
    openapi=OpenApiFunctionDefinition(
        name="get_weather",
        spec=openapi_weather,     # dict loaded from JSON/YAML file
        description="Get weather data",
        auth=OpenApiAnonymousAuthDetails(),
    )
)
```

## Supported auth types: `anonymous`, `API key`, `managed identity`

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# OpenAPI Authentication and Connections

***[EXAM]***

When the API needs a key on every call, declare an **API-key security scheme** in the OpenAPI spec so the platform attaches it automatically:

```
from azure.ai.projects.models import OpenApiConnectionAuthDetails

auth = OpenApiConnectionAuthDetails(connection_id=project_connection.id)
```

| Approach | Result |
|---|---|
| **API-key security scheme** | Key injected on every request — defined once |
| **Project connection** | Reuses the key already stored in the project connection |
| Adding the header manually | Repeats secret handling in code and risks leaking the key |
| Bearer/OAuth scheme | Expects an OAuth token, not an API key |
| Identity passthrough | Forwards the **caller's identity**, not the stored API key |

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# Project Connections — One Credential, Many Agents

***[EXAM]***

**Problem:** every agent and every application configures the Azure AI Search key for itself.

**Solution:** add a **connection** to the Search resource in the Foundry project, then reference that connection from each agent and app.

| Approach | Result |
|---|---|
| **Project connection to Azure AI Search** | One place to store, rotate, and audit the credential |
| A custom HTTP connection per app | Duplicated configuration and administration |
| One managed identity per app | Duplicated role assignments, still configured per app |
| Calling Search directly from each app | The search configuration is never shared |

**Not credential management:** enabling RBAC and disabling key-based access **hardens** the resource; a managed private endpoint controls **network** access. Neither centralizes the credential.

**Isolation still applies:** give each agent its **own identity** for tool and data access, so permissions can be granted, audited, and revoked per agent.



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# MCP Tools

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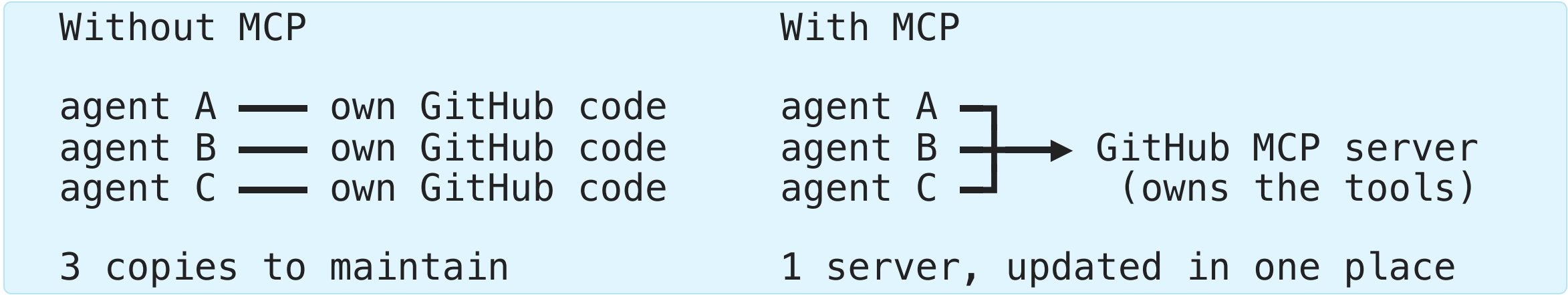
# Why MCP Exists

**Problem:** every tool you hand-code lives inside one agent. Ten agents that need the same GitHub tools means ten copies to build, secure, and update.




**MCP = Model Context Protocol.** A standard way for a *server* to publish tools and for any agent to consume them.

**Mental model:** MCP is a power socket. The agent brings the plug; it does not need to know how the power station works.

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# MCPTool — Remote MCP Servers

***[EXAM]***

**No manual session/client management needed when using built-in `MCPTool`:**

```
from azure.ai.projects.models import MCPTool, PromptAgentDefinition

mcp_tool = MCPTool(
    server_label="GitHub",
    server_url="https://api.githubcopilot.com/mcp/",
    allowed_tools=["list_repos"],    # optional: restrict accessible tools
    require_approval="never"         # or "always"
)
mcp_tool.update_headers({"Authorization": "Bearer {token}"})

agent = project_client.agents.create_version(
    name="github-agent",
    definition=PromptAgentDefinition(
        model="gpt-4.1",
        instructions="Help with GitHub tasks.",
        tools=[mcp_tool],
    )
)
```

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# MCP Tool Approval

***[EXAM]***

`require_approval="always"` → agent response contains `mcp_approval_request`

```
# Check for approval request in the response
if response_item.type == "mcp_approval_request":
    # Present to user or auto-approve
    await respond_with_approval(
        approval_request_id=response_item.id,
        approve=True
    )
```

**Key advantages of MCP vs. hard-coded tools:**

- **"Integrate once"** — tools updated on server; no agent redeployment needed
- **LLM-agnostic** — switch underlying model without reworking integrations
- **Standardized auth** — consistent authentication mechanism across tools

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# Foundry IQ — Knowledge

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# Retrieval Practice — Lab 3

Pause here and complete Lab 3 before continuing.

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# Controlling Tool Use — `tool_choice`

***[EXAM]***

By default the model decides. When an agent sometimes answers **without** calling its knowledge tool, the citations are ungrounded — force the call.

| Value | Behavior |
|---|---|
| `tool_choice="auto"` | Default — the model may skip every tool |
| `tool_choice="required"` | Must call **at least one** tool before answering |
| `tool_choice={"type": "bing_grounding"}` | Must call **that specific** tool |
| `tool_choice="none"` | May not call any tool |

```
response = client.responses.create(
    model=DEPLOYMENT,
    input=messages,
    tools=[mcp_tool],
    tool_choice="required",
)
```

**Required vs specific:** `"required"` guarantees *a* tool runs; it does not say *which*. To restrict one run to public web results only, name the tool: `{"type": "bing_grounding"}`. Naming `{"type": "azure_ai-search"}` would instead force the private index.

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# What is Foundry IQ?

***[EXAM]***

**Microsoft's managed knowledge platform for AI agents, built on Azure AI Search**

- Multiple agents share the same knowledge bases
- Uses MCP internally to connect agents to knowledge bases

**RAG steps:** Retrieve → Augment → Generate

**Supported data sources:**

| Data Source | Access type | Best for |
|---|---|---|
| Azure AI Search Index | Indexed | Existing search investment |
| Azure Blob | | Files (PDF, DOCX, TXT, MD, |

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# Classic RAG vs Agentic RAG

***[EXAM]***

**Classic RAG** searches once with the user's words, then answers.

```
question ---> one search ---> generate
```

**Agentic RAG** gives retrieval to the model as a tool, so it plans the search itself.

```
question ---> rewrite / split into sub-questions
                  |  searches run in parallel
                  v
            read results ---> not enough? search again ---> generate
```

| Choose | When |
|---|---|
| **Classic RAG** | One index, FAQ-style questions, fixed cost and latency |
| **Agentic RAG** | Answer needs **several chunks**, earlier turns must steer retrieval, or sources must be searched in parallel |

**Not the same as:** *iterative retrieval* (repeats a query without conversation-aware planning) or *chain of thought* (structures reasoning, not retrieval).

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# Connecting an Agent to a Knowledge Base

***[EXAM]***

```
from azure.ai.projects.models import MCPTool, PromptAgentDefinition

knowledge_tool = MCPTool(
    server_label="product-docs",
    server_url=f"{search_endpoint}/knowledgebases/product-documentation/mcp"
)

agent = project_client.agents.create_version(
    agent_name="product-support-agent",
    definition=PromptAgentDefinition(
        model="gpt-4o-mini",
        instructions="Answer product questions. Always cite your sources.",
        tools=[knowledge_tool]
    )
)
```

The Search credential comes from the **project connection**, not from application code.



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# Retrieval Instructions Pattern

## Three critical behaviors to specify:

```
retrieval_instructions = """
CRITICAL RULES:
- You must ALWAYS search the knowledge base before answering any question
- You must NEVER answer from your own knowledge or training data
- Every answer must include citations in this format:
   【doc_id:search_id†source_name】
- If the knowledge base doesn't contain the answer, respond with:
  "I don't have that information in my knowledge base."
"""
```

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# Retrieval Practice — Lab 4

Pause here and complete Lab 4, then check its solution.

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# Microsoft 365 Integration

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# Publishing an Agent to Teams/Copilot

***[EXAM]***

**Publishing creates an Agent Application with:**

- Dedicated invocation URL (stable across version updates)
- Agent identity (distinct Microsoft Entra app registration)
- Creates Azure Bot Service resource
- User data isolation

**Publishing scopes:**

| Scope | Visibility | Requires admin approval |
|---|---|---|
| **Shared** | "Your agents" in Teams | No — immediate |
| **Organization** | "Built by your org" in Teams | Yes |

**Required roles:** Azure AI Project Manager (publish), Azure AI User (invoke)

`Microsoft.BotService` provider must be registered in the subscription

**Post-publish:** Reassign RBAC to the published agent's new identity for Azure resources (AI Search, Storage) — dev-time permissions do not transfer

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# Microsoft 365 Publishing Options

**Direct Foundry publishing:**

- Creates an Agent Application with a stable invocation URL
- Creates an Azure Bot Service resource and Microsoft Entra application
- Generates a Microsoft 365 publishing package
- Routes new agent versions through the same public endpoint

**Other channels:**

| Channel | Use case |
|---|---|
| **Teams and Microsoft 365 Copilot** | Users work inside Microsoft 365 |
| **Web application preview** | Browser-based demonstrations and stakeholder testing |
| **Stable API endpoint** | Embed the agent in a custom application |
| **Azure Bot Service channels** | Slack, Telegram, Twilio, Facebook, and other |

**Use the Microsoft 365 Agents Toolkit** when the solution needs custom single sign-on, middleware, or multi-environment deployment.

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# Access Microsoft 365 Data with Work IQ

**Work IQ** gives an agent access to authorized workplace context such as:

- Emails and conversations
- Meetings and calendar information
- Documents and collaboration data

**Important boundaries:**

- The signed-in user's Microsoft 365 permissions still apply
- Request only the data needed for the task
- Explain which data sources the agent can use
- Validate citations and permissions before exposing results

**Work IQ is different from Foundry IQ:**

| | Foundry IQ | Work IQ |
|---|---|---|
| **Primary data** | Configured enterprise knowledge sources | Microsoft 365 workplace data |
| **Typical use** | Product, policy, and document knowledge | Personal and organizational work context |

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# Test and Operate a Published Agent

**Before publishing:**

1. Test normal, ambiguous, and adversarial prompts in the Foundry playground
2. Verify every configured tool and knowledge source
3. Check citations, authentication, and user-data isolation
4. Confirm the published identity has the required Azure RBAC roles

**After publishing:**

- Test in Teams and Microsoft 365 Copilot with representative users
- Monitor invocation failures, latency, tool calls, and user feedback
- Update the agent version without changing the stable invocation URL
- Reassign permissions whenever a published identity is newly created

**Common failure:** the agent works in Foundry but fails after publishing because development-time permissions do not transfer to the published identity.

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# Agent Workflows

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# Retrieval Practice — Lab 5

Pause here and complete Lab 5, then check its solution.

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# Workflow Patterns

***[EXAM]***

For a workflow that must pause before an external side effect, use an `ask_question` step and continue only when the returned value equals `"approved"`. This is a workflow-variable approval pattern, distinct from the SDK's typed `approve=True` tool-approval contract.

| Pattern | Description |
| --- | --- |
| **Sequential** | Fixed step-by-step pipeline; predictable, |

**Workflow Node Types:**

| Node | Sub-types / Purpose |
| --- | --- |
| **Invoke** | Calls an AI agent; returns free-text or structured JSON |

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# Choosing an Orchestration

***[EXAM]***

| Requirement | Choose |
|---|---|
| Deterministic steps, conditions, and shared state with minimal development effort | **A workflow** |
| Ordered work with approval before a risky update | **Sequential template + an Ask a question node** |
| Agents decide who contributes next | Group chat |
| Independent steps that can run at the same time | Concurrent |

**Why not the alternatives:** a multi-agent group chat is built for autonomous conversational collaboration and lets the agents choose the order; concurrent orchestration runs steps in parallel; threads and runs without a workflow — or separate agent runs coordinated in your own application code — mean you write and maintain the orchestration yourself.

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# Power Fx Formulas in Workflows

***[EXAM]***

**Logic in workflow nodes:**

```
```

**Variables:** `Local.*` (captured during execution), System variables (workflow context)

**Interpolation in message text:** wrap the expression in braces — `{Upper(Local.Var01)}`. Without braces the formula is printed as literal text.

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# Power Fx — variable scopes & operators

**Scope prefix is required** — bare `Var1` does not resolve.

| Prefix | Source |
|---|---|
| `Local.<name>` | Workflow variables you defined |
| `System.<name>` | Built-in (e.g. `System.RunId`) |
| `Trigger.<name>` | Fields from the triggering message / input |
| `Activity.<NodeName>.Output.<field>` | Output of an upstream node |

**Boolean operators:** `&&` `||` `Not(...)` `=` `<>` `>` `<` `>=` `<=`

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# If/Else node — "must have a value"

***[EXAM]***

The **If/Else node** takes a **boolean expression** (not a value-returning `If(...)`).

| Goal | Condition |
|---|---|
| Variable **must have a value** | `Not(IsBlank(Local.Var1))` |
| Variable is missing | `IsBlank(Local.Var1)` |
| List has at least one item | `Not(IsEmpty(Local.ItemList))` |
| Two values both required | `Not(IsBlank(Local.A)) && Not(IsBlank(Local.B))` |

**`IsBlank` vs `IsEmpty`:**

- `IsBlank(x)` — true when `x` is `null` **or** an empty string `""`
- `IsEmpty(coll)` — true when a **collection / table** has zero rows

> The function `If(cond, a, b)` **returns a value** — use it inside *Set Variable*, not as the condition of an If/Else node.

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# Invoke a Workflow from Code

```
conversation = openai_client.conversations.create()

stream = openai_client.responses.create(
    conversation=conversation.id,
    extra_body={"agent": {"name": "triage-workflow", "type": "agent_reference"}},
    input="Users can't reset their password from the mobile app.",
    stream=True,
)

for event in stream:
    if event.type == "response.completed":
        print(event.response.output_text)
```

**Check action status:** `event.item.type == ItemType.WORKFLOW_ACTION` → `event.item.status`

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# Microsoft Agent Framework

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# Framework Overview

**Open-source SDK; provider-agnostic** (Azure OpenAI, OpenAI, Anthropic, Copilot Studio, Foundry)

**vs. Foundry Agent Service:**

| | Microsoft Agent Framework | Foundry Agent Service |
|---|---|---|
| **Compute** | Your infrastructure | Fully managed |
| **Storage** | Your storage | Platform-managed |
| **Providers** | Any (OpenAI, Anthropic, …) | Foundry models |
| **Control** | Full code control | Configuration-based |

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# Core Components

| Component | Description |
| --- | --- |
| `Agent` | Main class: client + instructions + tools |
| `AzureOpenAIResponsesClient` | Connection + auth to Foundry project |
| `AgentSession` | Persistent conversation history (roles: USER, ASSISTANT, SYSTEM, TOOL) |
| **Built-in tools** | Code Interpreter, File Search, Web Search |
| `@tool` **decorator** | Marks Python functions as agent tools |

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# Custom Tools with @tool

```python
from microsoft_agent_framework import Agent, tool
from pydantic import Field
from typing import Annotated

@tool(name="get_stock_price", description="Get current price for a stock ticker")
def get_stock_price(
    ticker: Annotated[str, Field(description="Stock ticker symbol, e.g. MSFT")]
) -> float:
    return 428.50

agent = Agent(
    client=AzureOpenAIResponsesClient(...),
    instructions="You are a financial advisor.",
    tools=[get_stock_price]
)
session = AgentSession()
result = await agent.run("What is the price of MSFT?", session=session)
```

Use `Annotated` + `Field` from Pydantic for parameter descriptions

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# Multi-Agent Orchestration

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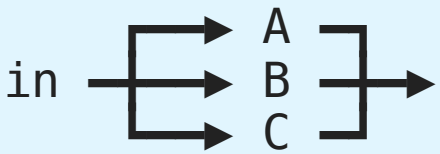
# Orchestration Patterns — the Shapes

Each pattern is really a picture of *who talks to whom, in what order*.

| | | |
|---|---|---|
| Sequential | A → B → C | each one refines the last |
| Concurrent |   | all at once, then combine |
| Group chat | A ↔ B ↔ C | a manager picks who speaks next |
| Handoff | A –?→ B or C | A decides who should take over |
| Magentic | plan → A → replan → C | plan first, adapt as it goes |

**Ask two questions to pick one:** Do the steps depend on each other? Is the order known before you start?



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# Orchestration Patterns

***[EXAM]***

| Pattern | Builder class | When to use | Avoid when |
|---|---|---|---|
| **Sequential** | `SequentialBuilder` | Step-by-step pipelines; progressive refinement | Stages can run in parallel |
| **Concurrent** | `ConcurrentBuilder` | Parallel, independent tasks; ensemble; voting | Tasks depend on each other |
| **Group chat** | `GroupChatBuilder` | Collaborative conversation; maker-checker; human-in-the-loop | Simple linear tasks; speed critical |
| **Handoff** | `WorkflowBuilder` + switch-case | Dynamic delegation; expertise unknown upfront | Fixed agent order known |
| **Magentic** | `MagenticBuilder` | Complex open-ended; plan-first; dynamic task ledger | Simple/deterministic tasks |

**All patterns:** `participants()` → `build()` → `run()` or `run_stream()`
**Results:** `get_outputs()` → iterate messages with `.author` and `.content`

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# Workflow Edge Types

| Edge | Description |
|---|---|
| **Direct** | Sequential: A → B |
| **Conditional** | Execute only when condition is met |
| **Switch-Case** | Route to different executors based on predefined cases |
| **Fan-Out** | Send one message to multiple executors simultaneously |
| **Fan-In** | Combine outputs from multiple executors into one |

**Built-in events:** `WorkflowStartedEvent`, `WorkflowOutputEvent`, `WorkflowErrorEvent`, `ExecutorInvokeEvent`, `ExecutorCompleteEvent`, `RequestInfoEvent`

Workflows support **checkpointing** (save/resume state)



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# A2A Protocol

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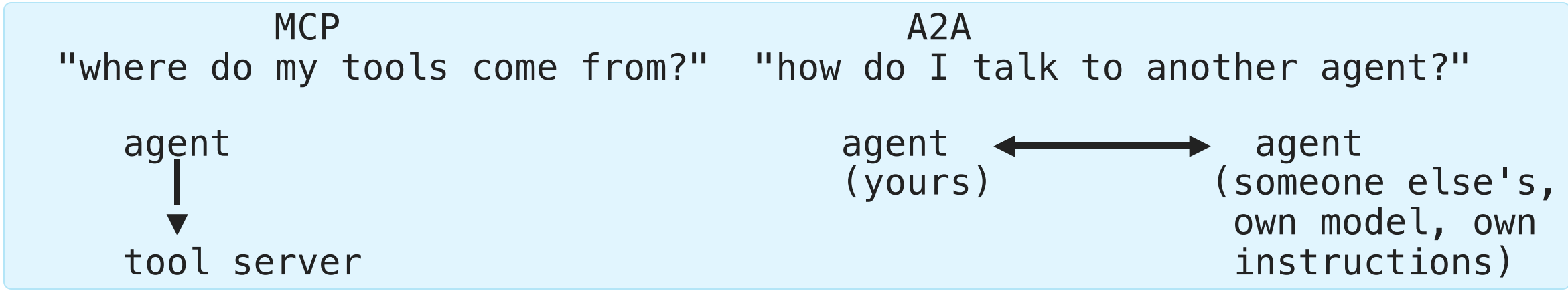
# MCP and A2A Are Not Competitors

They solve two different questions.




| Question | MCP | A2A |
|---|---|---|
| Other side is… | a tool provider | another autonomous agent |
| Who picks the model? | you, for your agent | each agent picks its own |
| Discovery | server URL + tool list | `agent-card.json` |

**One sentence:** MCP gives an agent *hands*; A2A gives it *colleagues*.

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# Agent-to-Agent (A2A)

**Standardized framework for agent discovery, communication, and coordinated task execution**

**Key advantage over MCP:** Each A2A agent chooses its own LLM; integrated authentication; cross-vendor

**Agent Card** — discovery document at `/.well-known/agent-card.json`:

```
{
  "name": "weather-agent",
  "description": "Provides weather forecasts",
  "endpoint_url": "https://my-agent.example.com/",
  "capabilities": { "streaming": true },
  "skills": [
    {
      "id": "weather-forecast",
      "name": "Weather Forecast",
      "description": "Get weather for a location",
      "tags": ["weather", "forecast"],
      "examples": ["What is the weather in Paris?"]
    }
```

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# A2A Server and Client

**Server components:**

| Component | Role |
|---|---|
| **Agent Card** | Describes capabilities; at `/.well-known/agent-card.json` |
| **Request Handler** | Routes requests to `AgentExecutor`; manages Task Store |
| **Server Application** | Starlette + Uvicorn (ASGI); listens on HTTP |

**AgentExecutor interface:**

- `execute` — processes requests; sends results via `EventQueue`
- `cancel` — handles task cancellation

**Client request types:**

- **Non-streaming** — send message, wait for complete response
- **Streaming** — incremental responses for long-running tasks



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# Advanced Generative AI and Agent Operations

Operate agents by separating runtime state, tools, evaluation, and monitoring.

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# Agent Lifecycle and Runtime State

***[EXAM]***

**Agent:** instructions, model, and tools

**Thread or conversation:** persistent context

**Message:** user, assistant, or tool content

**Run:** one execution against the conversation

Use a durable conversation or persisted message store when history must survive sessions.

**To keep a full case history across sessions, store the conversation ID and reuse it.** Saving only the final model response loses the tool calls and tool outputs; summarizing condenses history instead of restoring the complete execution record.



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# Agent Memory and Scopes

***[EXAM]***

**Memory is not RAG and not conversation history.**

| Feature | Stores |
|---|---|
| **Conversation** | The full turn history of one ongoing case |
| **Knowledge** | Source documents retrieved as evidence |

**Scopes decide who shares a memory:**

| Scope | Lifetime |
|---|---|
| `session` | Discarded when the session ends |
| `"{{$userId}}"` | Per signed-in user, across sessions — no ID in code |

```
memory_tool = MemoryTool(scope="{{$userId}}")

agent = project_client.agents.create_version(
    name="support-agent",
    definition=PromptAgentDefinition(model="gpt-4.1", tools=[memory_tool]),
)
```

The `{{$userId}}` placeholder is substituted at runtime, so each user is isolated without generating identifiers yourself.

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# Evaluate, Control, and Monitor Agents

***[EXAM]***

- Use a groundedness gate or evaluator to stop unsupported answers
- Use `Search Index Data Reader` for read-only Search access
- Use reflection to regenerate incomplete answers
- Treat `approve=True` as a typed approval decision, not a string
- Monitor tool calls, latency, tokens, handoffs, quality, safety, and versions

**Error analysis:** inspect retrieval, prompt, model, tool, orchestration, safety, and user-experience layers in order.

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# Evaluate an Agent in a Pipeline

***[EXAM]***

**Which evaluators for a RAG agent?**

| Evaluator | Answers |
|---|---|
| **Retrieval** | Did the search return the right documents? |
| **Groundedness** | Does the answer follow those documents? |
| Fluency and coherence | Readability and logical flow — not correctness |

**Gate a pull request with GitHub Actions:**

- Authenticate with **OIDC** — federated, no long-lived stored credential
- Set the **project endpoint** so the workflow knows which Foundry project to evaluate against
- **Fail the workflow** when evaluation fails, so an unsafe version cannot merge

The evaluation config path, the model deployment name, and the tenant ID do not identify the project.

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# Key Takeaways

1. **Agent types:** Declarative (prompt-based / workflow) + Hosted
2. **Custom tools:** Function calling, Azure Functions, OpenAPI 3.0, Logic Apps
3. **MCP:** `MCPTool` attaches remote MCP servers; `require_approval` controls autonomy
4. **Foundry IQ:** Managed RAG platform; multiple data sources; citation pattern
5. **Agent Framework:** Open-source, provider-agnostic; `@tool` decorator for function tools
6. **Orchestration patterns:** Sequential, Concurrent, Group chat, Handoff, Magentic
7. **A2A:** `/.well-known/agent-card.json` for discovery; Agent Skills describe capabilities

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# Problem: "The Agent Gave a Bad Answer"

That sentence is not a bug report. It could mean any of these:

- The search returned the wrong document
- The instructions never said to cite sources
- The tool threw an error and the model guessed instead
- The model is too small for the task
- Everything worked — last week's version was simply different

**Need:** vocabulary precise enough to say *which part* failed.

This presentation is that vocabulary.

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# Retrieval Practice — Lab 6

Pause here and complete Lab 6, then check its solution.

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# Classic Agent Service Lifecycle

***[EXAM]***

For the exam, distinguish the persistent conversation objects from one execution:

```
Agent        = instructions + model + tools
Thread       = persistent conversation state
Message      = user, assistant, or tool content in a thread
Run          = one execution of an agent against a thread
```

**Typical flow:**

```
create agent
  → create thread
  → add user message
  → create and process run
  → inspect run status
  → list messages
```

**Why it matters:** a thread stores context, while a run represents the model and tool execution that produces the next response.

**Current API note:** newer Responses API and Agent Framework examples may use

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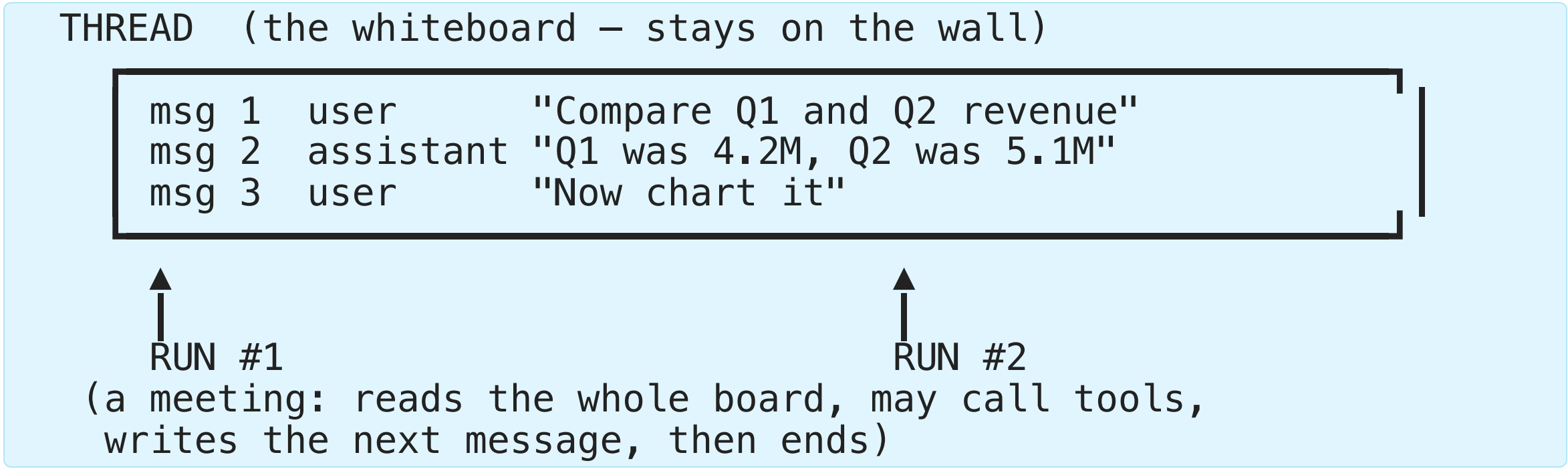
# Thread vs Run — the Whiteboard and the Meeting




**The thread persists; a run is over as soon as it produces its message.** Cost, latency, tool calls, and failures all belong to a *run*. Context belongs to the *thread*.

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# Retrieval Practice — Lab 7

Pause here and complete Lab 7, then check its solution.

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# Tool Taxonomy

***[EXAM]***

| Tool | Primary capability | Typical control |
|---|---|---|
| **Function tool** | Application-owned operation | Your schema, validation, and execution |
| **MCP tool** | Discoverable remote tools | Server permissions and approval policy |
| **File search** | Retrieval from uploaded files or vector stores | Vector store and citation controls |

**Selection rule:** use the narrowest tool that solves the task, and require approval for actions with external side effects.

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# Azure AI Search RBAC

***[EXAM]***

For read-only agent queries against a shared Azure AI Search service, assign the agent identity the **Search Index Data Reader** role. Managed identities let each agent receive a separate, auditable data-plane permission.

**Role boundaries:**

- **Search Index Data Reader:** query documents in indexes
- **Search Index Data Contributor:** read and push documents
- **Search Service Contributor:** manage indexes, indexers, and data sources

Use the narrowest data-plane role that satisfies the agent's task; do not grant service-management permissions to a retrieval-only agent.

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# Retrieval Practice — Lab 8

Pause here and complete Lab 8 before continuing.

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# Generation Controls

***[EXAM]***

| Parameter | Main effect | Exam consideration |
|---|---|---|
| `temperature` | Randomness and variation | Lower for repeatable structured output |
| `top_p` | Probability-mass vocabulary cutoff | Usually tune instead of temperature, not both |
| `max_tokens` / `max_output_tokens` | Output budget | Affects completeness, cost, and rate-limit reservation |
| `frequency_penalty` | Reduces repeated tokens | Useful when responses repeat phrases |
| `presence_penalty` | Encourages new topics | Useful when broader exploration is desired |

**Do not use a parameter to solve the wrong problem:** grounding fixes missing knowledge; a penalty does not fix hallucination, and temperature does not add facts.

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# Retrieval Practice — Lab 9

Pause here and complete Lab 9, then check its solution.

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# Reflection and Self-Critique Loops

***[EXAM]***

**A quality loop asks a second evaluator or agent to inspect the first answer.**

```
draft answer
  → check against requirements and source context
  → identify unsupported claims or missing steps
  → revise or reject
  → return answer with evidence
```

**Useful checks:**

- Is every claim grounded in retrieved context?
- Does the answer satisfy the requested format?
- Did the tool call use valid and authorized arguments?
- Should a human review this result?

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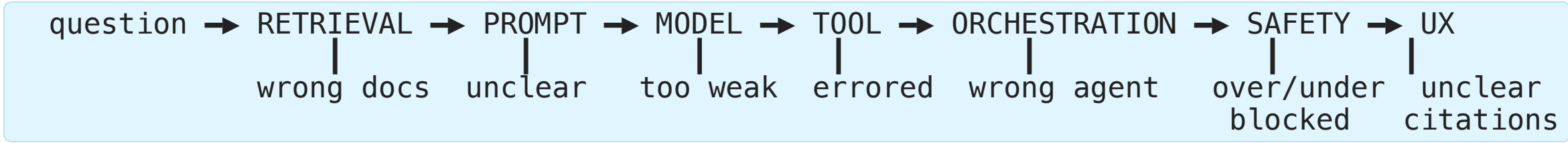
# Evaluation and Error Analysis

***[EXAM]***

A wrong answer has a *first* broken link. Walk the chain in order and stop at the first failure.




**Separate failures by layer:**

| Failure | Diagnostic question |
|---|---|
| Retrieval | Did the index return the right source and chunk? |
| | Were instructions, context, and examples |

Use a labeled evaluation dataset, reproduce the failure, change one layer, and compare the before/after metrics.

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# Production Agent Monitoring

***[EXAM]***

**Monitor more than HTTP success:**

- Run duration and time spent in each model or tool call
- Prompt, completion, and tool tokens
- Rate-limit responses and retries
- Tool-call validity, failures, and approval requests
- Groundedness, relevance, coherence, fluency, and safety scores
- Handoff routes, loop counts, and abandoned runs
- Model, prompt, agent, and knowledge-source versions

**Alert on:** quality regression, rising latency, cost spikes, repeated tool errors, unsafe outputs, and retrieval failures.

**MSLearn status:** tracing, evaluation, and user feedback are covered; this complete operational view is primarily an exam synthesis.



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# Key Takeaways

1. **Thread:** persistent context; **run:** one execution; **message:** content within the thread
2. **Tools:** choose the narrowest safe tool and control external side effects
3. **Parameters:** temperature, top P, token limits, frequency penalty, and presence penalty solve different problems
4. **Reflection:** draft, inspect, revise, and measure the added cost
5. **Error analysis:** isolate retrieval, prompt, model, tool, orchestration, safety, and UX failures
6. **Operations:** monitor tokens, latency, tools, quality, safety, versions, and retries
7. **MSLearn relationship:** the building blocks are present, but this exam-oriented lifecycle and operations synthesis is not a

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# Human Approval Contract

***[EXAM]***

When a tool call requires approval, the SDK submits a Boolean decision such as `approve=True` or `approve=False` tied to the tool call. The string `"approved"` is not the SDK approval value.

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# Retrieval Practice — Lab 10

Pause here and complete Lab 10, then check its solution.

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# Appendix

Relationship to the Microsoft Learn path

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# Coverage Map

| Exam objective | MSLearn status |
| --- | --- |
| Agent `thread` / `run` / `message` lifecycle | **Indirect** — Agent Framework sessions and workflow runs are covered; the classic lifecycle is not central |
| Function, MCP, file search, code interpreter, and Bing grounding tools | **Partial** — most tools are covered; the exam grouping is not presented as one comparison |
| Prompt parameters and generation controls | **Partial** — temperature, top P, and token limits are covered; frequency and presence penalties are not |
| Reflection, self-critique, and error analysis | **Indirect** — evaluation is covered; these patterns are not developed deeply |
| Agent production monitoring | **Indirect** — tracing and feedback appear, but the exam checklist is broader |



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